PSYCHOLOGISCHES EXPERIMENTALPRAKTIKUM

SOSE 2025

Lecturer: Lingyue Chen

TECHNISCHE UNIVERSITÄT DARMSTADT

Centre for Cognitive Science

Sensorimotor Control & Learning

Week	Date	Topics
1	24.04	Project pitches and group assignments
2	29.04	Organization & preparation
3, 4	06.05, 13.05	Research question, study design, grouping
4 – 6	14.05 – 30.05	Experiment coding
7 – 9	02.06 – 20.06	Data collection
10, 11	23.06 – 04.07	Data analysis & interpretation
12, 13	07.07 – 18.07	Poster preparation
14	25.07	Poster session
19	28.08	Written report submission

Take Home Reading Qs

Task

What is target-pointing task?

What about reaching & avoiding task?

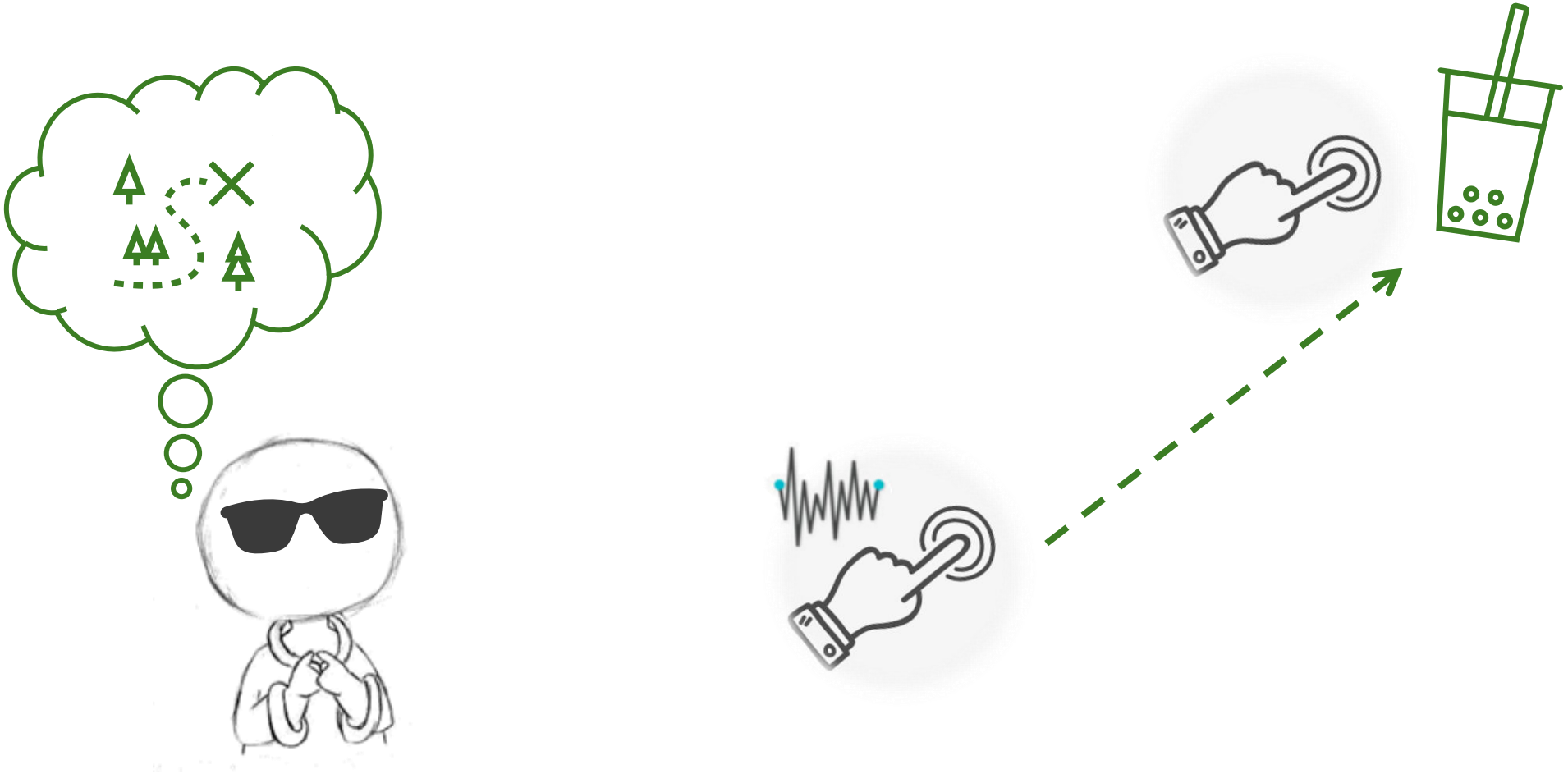
Sensorimotor mapping

What does map between sensory stimuli & target distance mean?

Pre- / Post-Test & Training

How to train & test participants' performances

Vibrotactile Cues in Spatial Perception

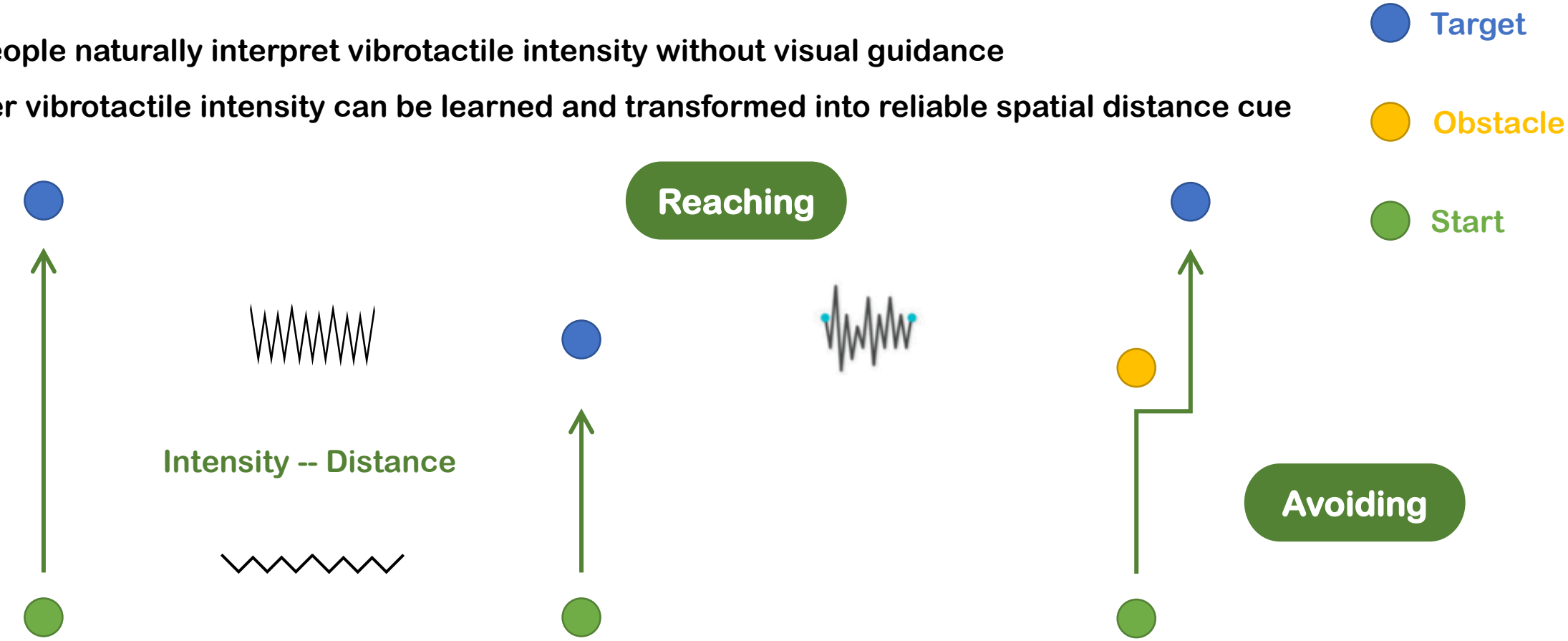


Task

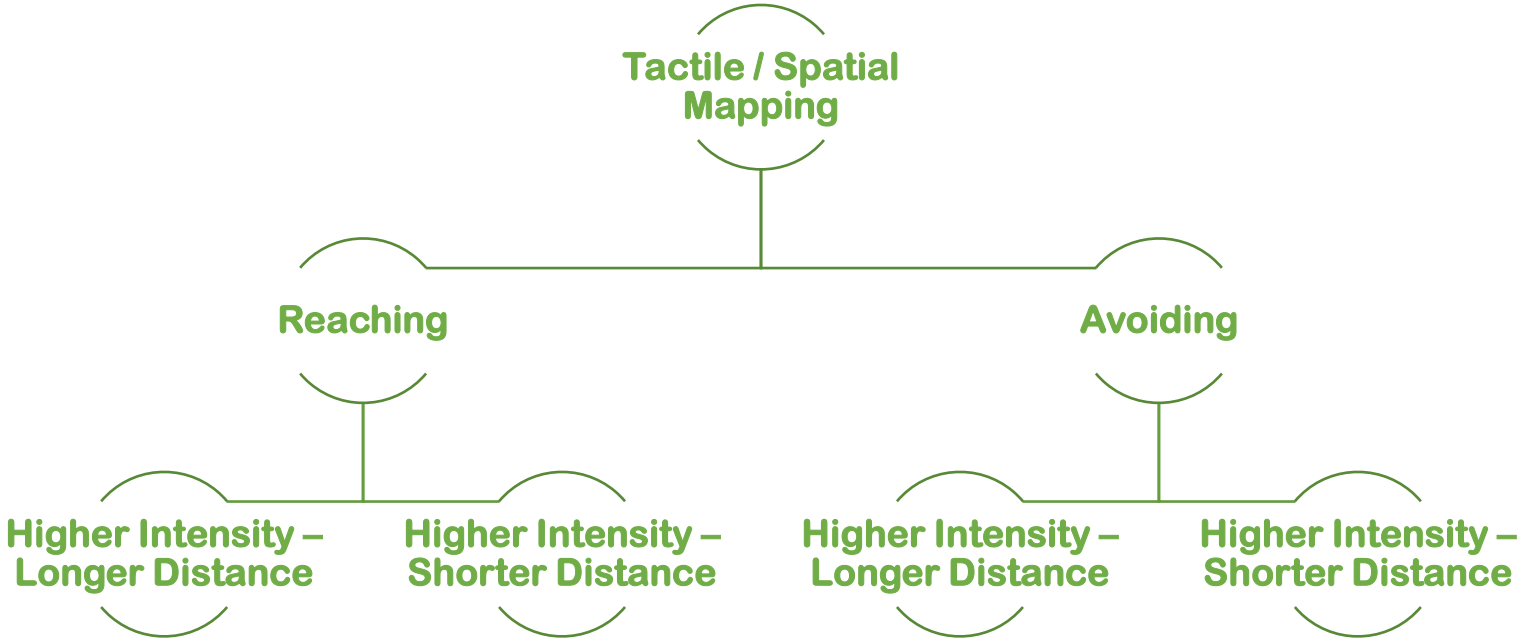
Sensorimotor mapping

Vibrotactile Intensity → Spatial distance

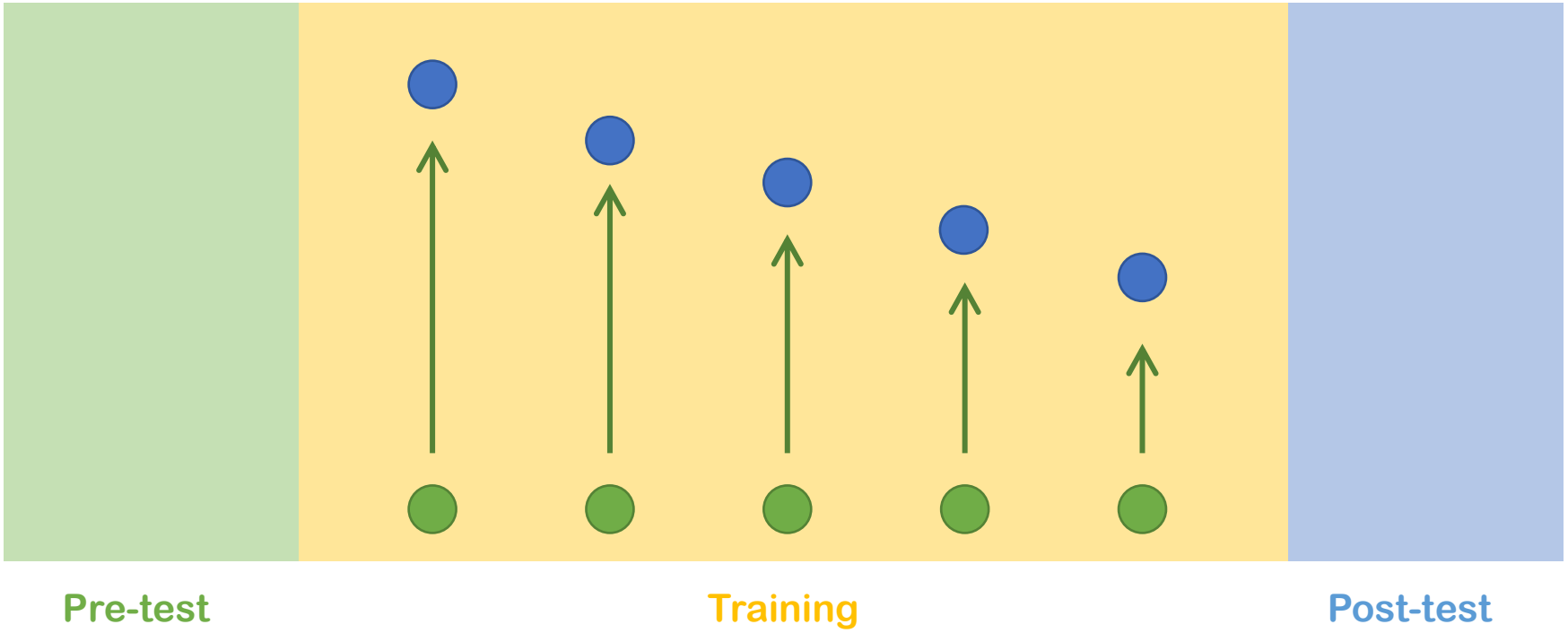
- How people naturally interpret vibrotactile intensity without visual guidance
- Whether vibrotactile intensity can be learned and transformed into reliable spatial distance cue



Study Design



Study Design

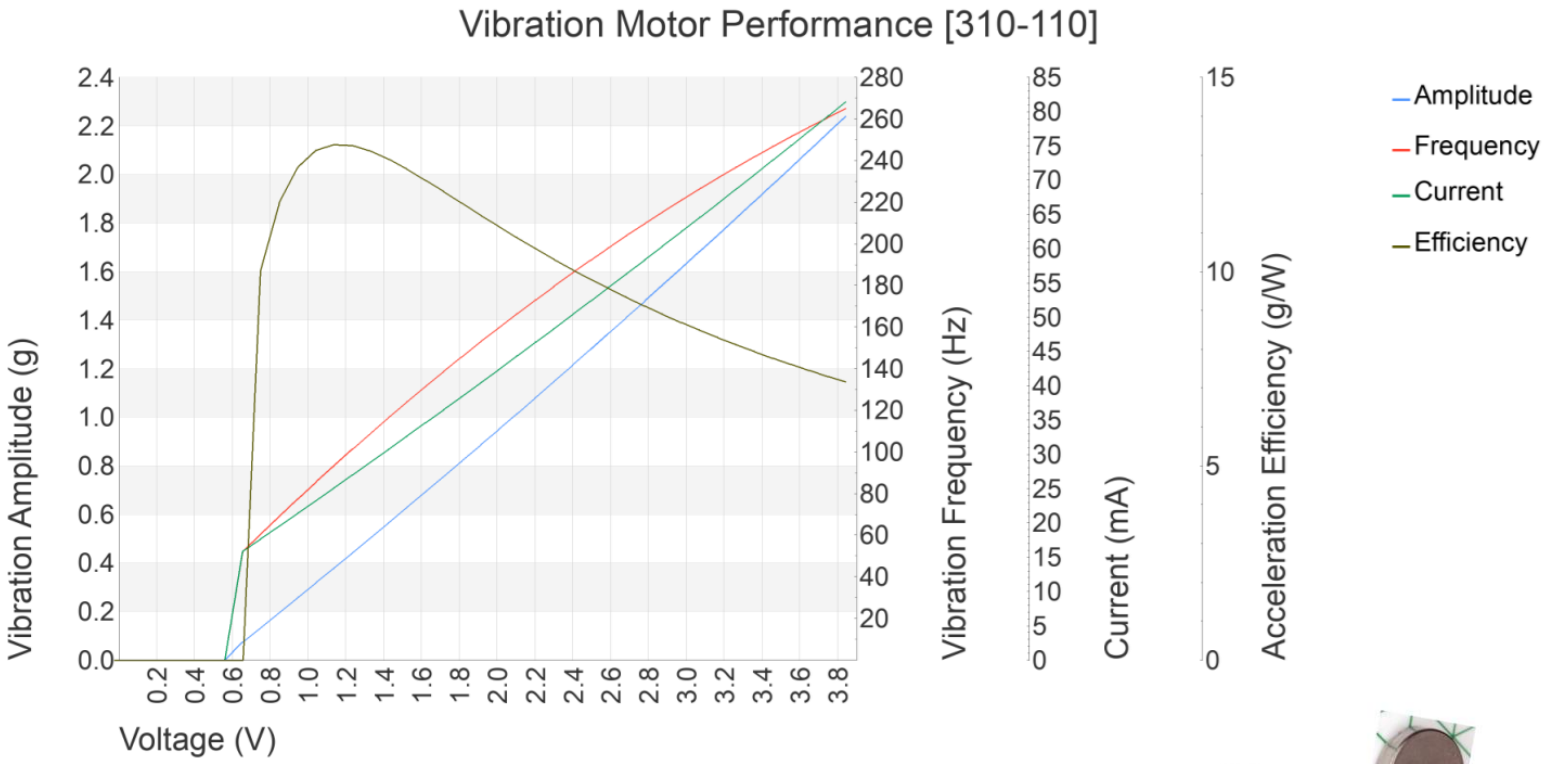


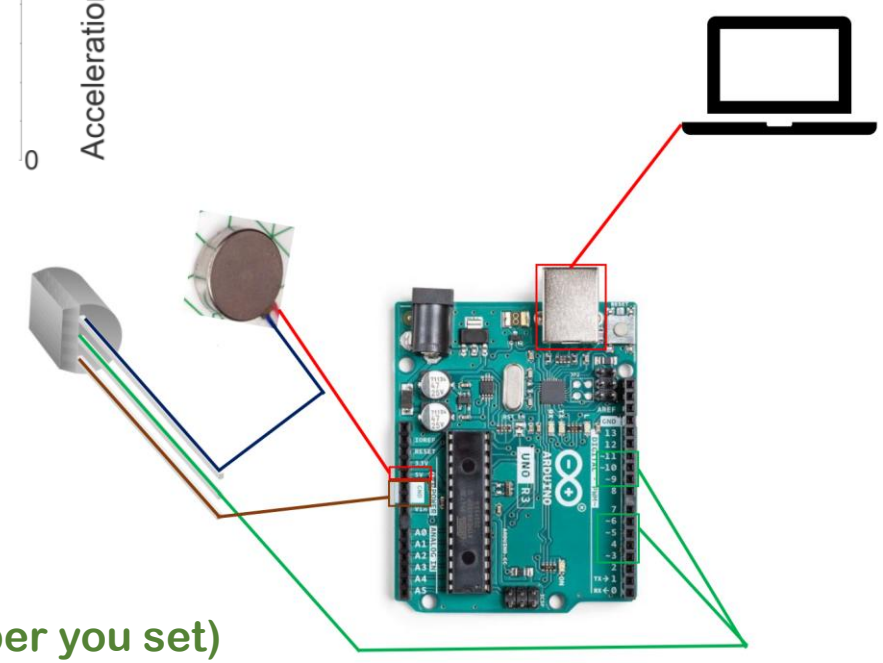
Study Design

Group 1	Avoiding	Reaching	Reaching	Reaching	Avoiding	Reaching
Group 2	Reaching	Avoiding	Avoiding	Avoiding	Reaching	Avoiding
Block No	1	1	5	1	1	1
Target Distance No						
Tactile intensity Level						
Target Repetition No						
Total Trial No / Block						
Total Trial No						
	Pre-test		Training	Post-test		Mini-Post

TYPICAL DC MOTOR PERFORMANCE CHARACTERISTICS

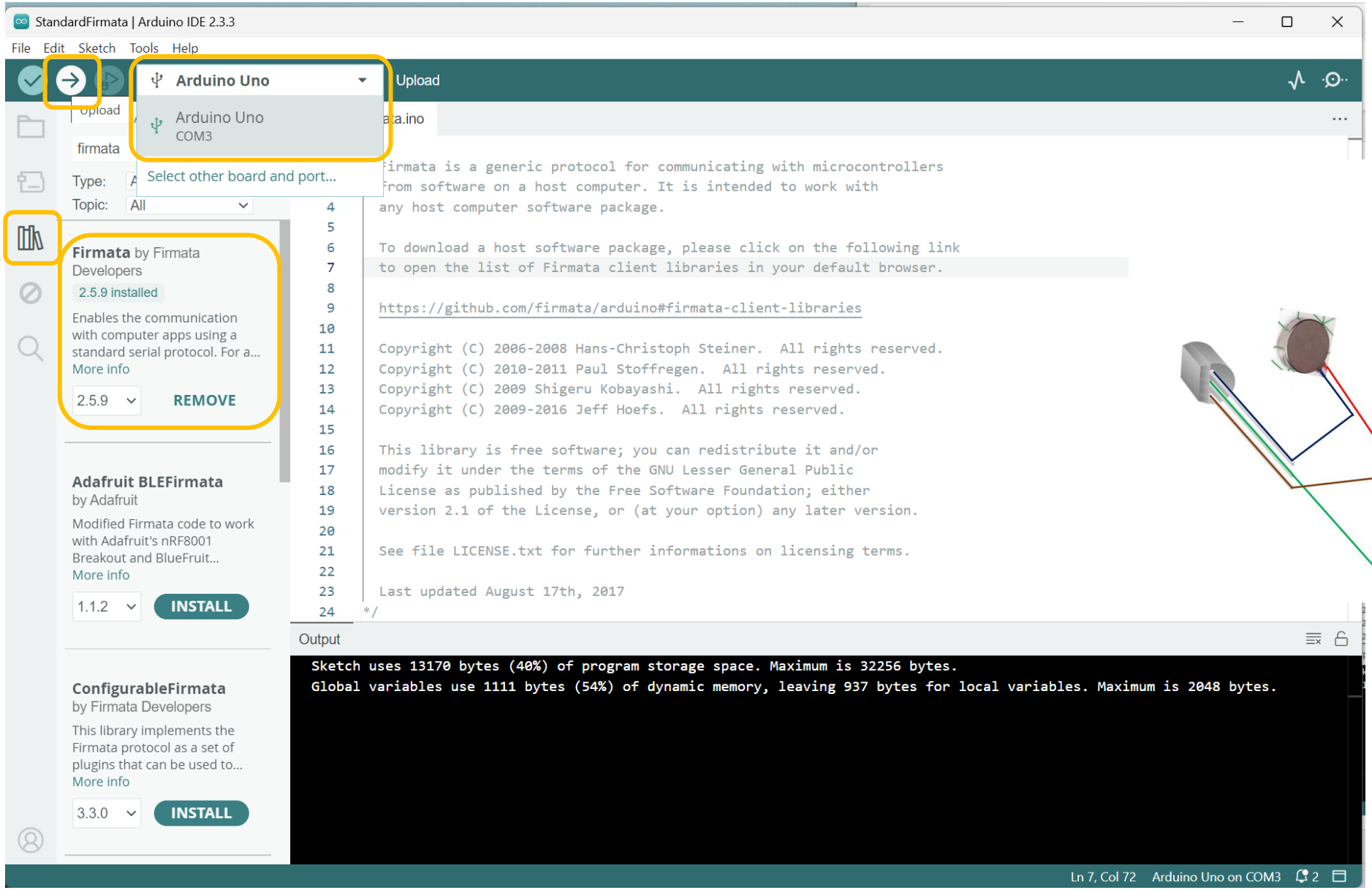
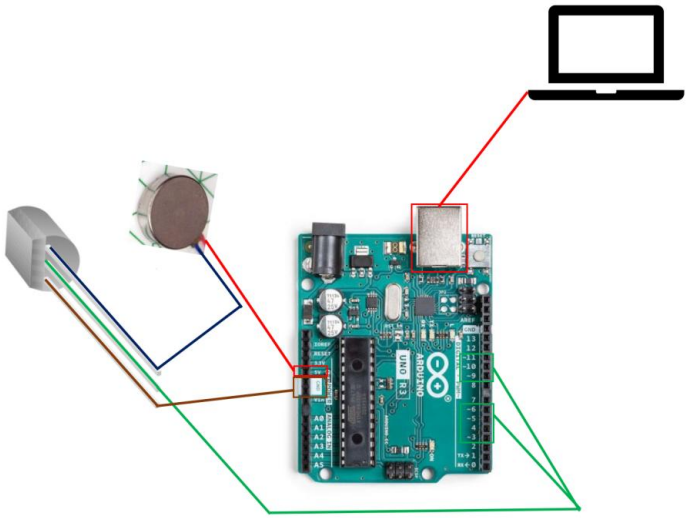
Arduino





3 cables
Power: 5V
Grand: GND
Pin: 10 (the pin number you set)

Arduino


StandardFirmata | Arduino IDE 2.3.3
File Edit Sketch Tools Help
Arduino Uno
Arduino Uno
COM3
Select other board and port...
Upload
firmata
Type:
Topic: All
Firmata by Firmata Developers
2.5.9 installed
Enables the communication with computer apps using a standard serial protocol. For a...
More info
2.5.9
REMOVE
Adafruit BLEFirmata
by Adafruit
Modified Firmata code to work with Adafruit's nRF8001 Breakout and BlueFruit...
More info
1.1.2
INSTALL
ConfigurableFirmata
by Firmata Developers
This library implements the Firmata protocol as a set of plugins that can be used to...
More info
3.3.0
INSTALL
Firmata is a generic protocol for communicating with microcontrollers
from software on a host computer. It is intended to work with
any host computer software package.
To download a host software package, please click on the following link
to open the list of Firmata client libraries in your default browser.
https://github.com/firmata/arduino#firmata-client-libraries
Copyright (C) 2006-2008 Hans-Christoph Steiner. All rights reserved.
Copyright (C) 2010-2011 Paul Stoffregen. All rights reserved.
Copyright (C) 2009 Shigeru Kobayashi. All rights reserved.
Copyright (C) 2009-2016 Jeff Hoefs. All rights reserved.
This library is free software; you can redistribute it and/or
modify it under the terms of the GNU Lesser General Public
License as published by the Free Software Foundation; either
version 2.1 of the License, or (at your option) any later version.
See file LICENSE.txt for further informations on licensing terms.
Last updated August 17th, 2017
*/
Output
Sketch uses 13170 bytes (40%) of program storage space. Maximum is 32256 bytes.
Global variables use 1111 bytes (54%) of dynamic memory, leaving 937 bytes for local variables. Maximum is 2048 bytes.
Ln 7, Col 72
Arduino Uno on COM3